Shengmiao (Samuel) Jin

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Education

Cornell University August 2025 - May 2030 *(expected)*
Ph.D. Student in Robotics
- Minor in Computer Science

University of Illinois Urbana-Champaign August 2021 - May 2025
Bachelor of Science in Electrical Engineering with Highest Honors *GPA: 3.85/4.0*
- Minor in Computer Science Minor GPA: 4.0/4.0

Honors and Awards

NSF Graduate Research Fellowship April 2026

Best Entertainment and Amusement Papers Finalists IROS 24

James Scholar Spring 2022 - Spring 2024

Dean's List Fall 2021, Spring 2022, Fall 2023, Spring 2024

Valedictorian, West Nottingham Academy Class of 2021 May 2021

Research Experience

Cornell EmPRISE Lab + Praxis Lab August 2025 - Present
Advisor: Prof. Tapomayukh Bhattacharjee and Prof. Preston Culbertson
- Worked on generalized tactile representation learning methods that enable manipulation policies to transfer across heterogeneous tactile sensor types without requiring retraining.
- Investigating efficient algorithms for continual learning at deployment, targeting online adaptation, sim-to-real transfer, and multimodal learning in robotic manipulation.
- Studying the role of infrastructural-level support in enabling tactile-based policies, and formalizing the utility of tactile sensing for contact-rich manipulation tasks using value-of-information theory.

UIUC RoboTouch Lab September 2023 - May 2025
Advisor: Prof. Wenzhen Yuan
- Designed and implemented an algorithm that decomposes an arbitrary image into a robot trajectory. Experimented and designed an algorithm for the robot to be able to handle liquid in an even manner. Integrated the overall pipeline presented and co-first-authored a paper published in IROS 24. [4]
- Led a project on estimating the mean and uncertainty of the center of mass for any arbitrary object with Active Perception. Submitted a first-author paper to ICRA 25. [1]
- Designed a learning approach to get sensor-invariant tactile image representations with calibration images and a sim2real pipeline. Compared and outperformed some baseline SOTA on several downstream tasks. [3]
- Investigating high-resolution tactile representation for dexterous in-hand object re-position. Training the policy on IssacGym with a teacher-student network, and zero-shot transferring to the real LEAP hand.

Teaching Experience

Cornell | *Graduate Teaching Assistant*
- CS 5757 - Optimization Methods for Robotics Spring 2026
- CS 4750/5750 - Foundations of Robotics Fall 2025

UIUC | *Undergraduate Grader*
- ECE 311 - Digital Signal Processing Lab Spring 2024
- ECE 210 - Analog Signal Processing Fall 2023

Publications

[1] **Shengmiao Jin**, Yuchen Mo, and Wenzhen Yuan, "Learning to double guess: An active perception approach for estimating the center of mass of arbitrary objects," *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[2] Zhi Wang*, Yuchen Mo*, **Shengmiao Jin**, and Wenzhen Yuan, "Doorbot: Closed-loop task planning and manipulation for door opening in the wild with haptic feedback," *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[3] Harsh Gupta*, Yuchen Mo*, **Shengmiao Jin**, and Wenzhen Yuan, "Sensor-invariant tactile representation," *International Conference on Learning Representations (ICLR)*, 2025.

[4] Xinyuan Luo*, **Shengmiao Jin***, Hung-Jui Huang, and Wenzhen Yuan, "An intelligent robotic system for perceptive pancake batter stirring and precise pouring," *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

* Equal Contribution

Other Projects and Activities

ASAP-Style Sim2Real Transfer for Dexterous Manipulation | CS 6750 Final Project — Fall 2025

- Implemented a ASAP-style sim2real transfer algorithm to transfer an Isaac-trained policy to Mujoco with a delta action module.

Phantom Navigation | ECE 598 CS Final Project — Spring 2025

- Implemented a Phantom Touch-based navigation grove with haptic, prototyped the hardware and design the algorithm for navigation.

Robot Drawing | ECE 470 Final Project — Fall 2024

- Implemented a Minimum Spanning Tree plus Edge Detection algorithm to decompose any drawing into a robot trajectory so that we can use the robot to draw an arbitrary input image.

End-to-End Autonomous Vehicle Driving with Imitation Learning | ECE 484 Final Project — Spring 2024

- Implemented and trained Imitation Learning algorithm with data collected on our own at Highbay-IRL. Tested the algorithm on the GEM vehicle. The final video can be found on Youtube.

Artwork Style Classification and Tranfer | CS 445 Final Project — Fall 2023

- Implemented and trained an EfficientNetV2-based classification model, a VGG-based Neural Style transfer, and a CycleGAN-based style generator.

Reaction Wheel Pendulum | ECE 486 Final Project — Fall 2023

- Implemented a 3-state PD controller with Decoupled Observer and friction compensation that can allow a pendulum to reject disturbance and stay at an unstable equilibrium position with only a rotor

Tetris On FPGA | ECE 385 Final Project — Summer 2023

- Implemented the classic game Tetris on an Intel MAX10 FPGA, programmed using SystemVerilog for hardware level and C for software level.

Computer Vision Team Member | Illini Robomaster — Fall 2022

- Part of Illini Robomaster CV Team that implemented object detection and tracking algorithm for competition

BitCorn | ECE 220 Honors Project — Spring 2022

- Implemented a Bitcoin Price prediction model using LSTM and updated daily results on a website